

GRiddles Submission: Evolving Maze

Andrew Smith

andysmith9433@gmail.com

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Problem Statement

A 10×10 grid is divided into 100 unit squares and has a wall around its perimeter. A pawn starts at the square (1, 1).

Two players, the Wall-Setter and the Pawn-Pusher, play a game, taking alternate turns. The Wall-Setter takes the first turn. On the Wall-Setter's turn, they can place any (possibly zero) number of walls on the grid lines between touching squares. However, they can never place walls in a way that completely blocks the pawn from reaching the square (10, 10). On the Pawn-Pusher's turn, they move the pawn exactly one step up, down, left or right to an adjacent square, as long as a wall does not block the move.

The game ends when the pawn reaches the square (10, 10). The final score of the game is the total number of moves the Pawn-Pusher made. The Wall-Setter wants to make the score as high as possible and the Pawn-Pusher wants to make the score as low as possible.

What strategies would you suggest for them?

Solution

1. Definition of Terms

Before diving into the solution, I will define a number of terms that correspond to different states that squares in the maze can be in.

1. **Branch Points:** I define these as squares in the maze where more than 2 sides can be traversed.
2. **Exit Points:** I define exit points as a particular class of branch points where all possible sides are traversable, all but one of the adjacent squares are also exit points, **and** the square (10, 10) is reachable from these exit points.
3. **Branch:** I define branches as corridors in the maze where exactly 2 sides are traversable.
4. **Dead End:** I define dead ends as squares in the maze at the end of branches with only one route out to a path that reaches (10, 10)

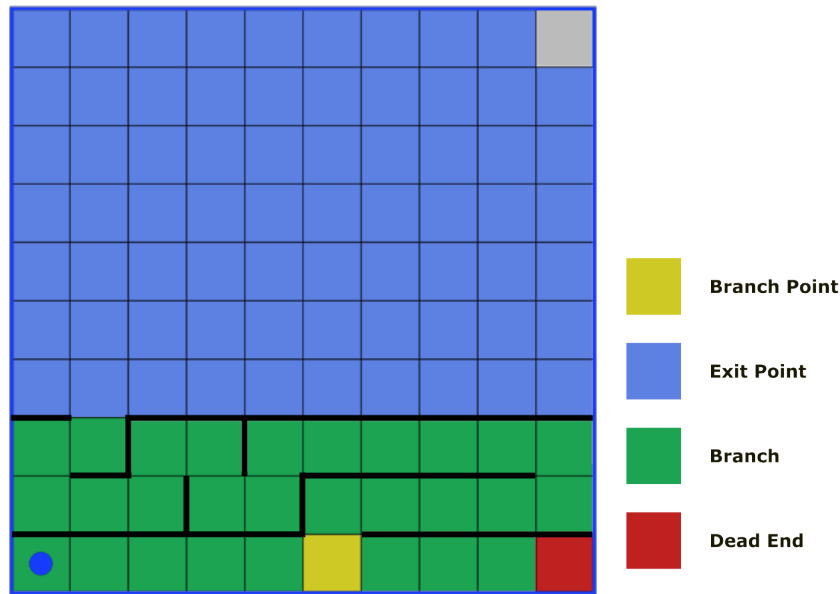


Figure 1: The Types of Grid Square in a Game

2. Pawn-Pusher Strategy

It is trivial to show that the Wall-Setter can **always** force the Pawn-Pusher to visit every square of the grid, simply imagine a pre-drawn maze which snakes around every square of the grid between the starting square and the final square which the Wall-Setter draws on turn 1. Such a maze is shown in Fig. 2. The Pawn-Pusher would be forced to move all the way to (9,9). They would then have to choose to move to either (9,10) or (10,9) at which point the Wall-Setter would block their access to (10,10) and force them to head around to get to (10,10).

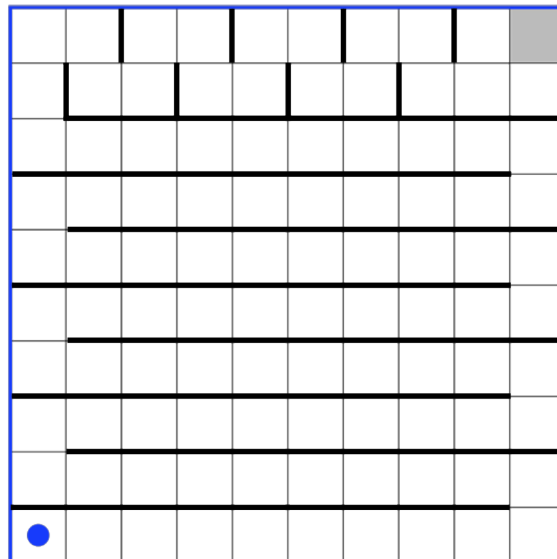


Figure 2: A set of walls the Wall-Setter could draw on turn 1 to force the Pawn-Pusher to visit every square.

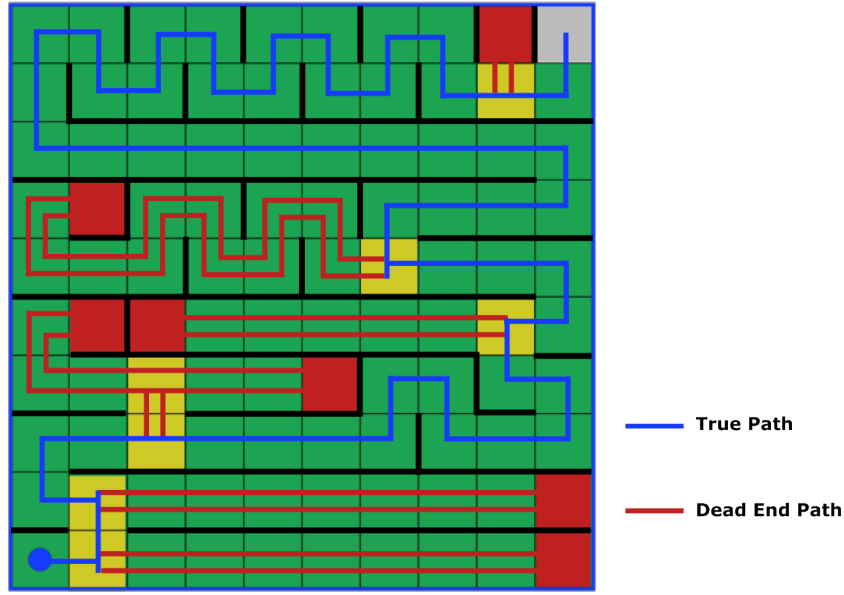


Figure 3: A set of dead end paths and the true path.

The Pawn-Pusher should therefore focus on a strategy which involves minimising the number of revisits to each square, since visiting every square at least once is unavoidable without a mistake from the Wall-Setter.

Each time the Pawn-Pusher reaches a branch point in the maze with more than 1 direction that is currently untraversed, they should choose the direction that has the shortest path to the final square (10, 10). They should continue along that path until they reach a dead end and can go no further, at which point they turn around, head back to the branch point, and continue along a different, untraversed path.

For every game, there will be a set of dead end paths that the Pawn-Pusher has traversed $B = \{b_1, b_2, \dots, b_N\}$ with lengths $L = \{l(b) | b \in B\}$, and one single "true" path from (1, 1) to (10, 10), l_T , that represents the actual fastest path taken by the Pawn-Pusher from (1, 1) to (10, 10) once all dead end branches have been removed. This is shown in Fig. 3. By traversing each path until it hits a dead end before turning back, the Pawn-Pusher ensures that each dead end is traversed exactly once.

During a traversal of a dead end branch the Pawn-Pusher will make $2l(b)$ steps. The final score of the game will therefore be,

$$s = l_T - 1 + 2 \times \sum_i l(b_i) \quad (1)$$

Where the -1 comes from the fact I need to remove the starting square (1, 1) from the true path l_T when considering its overall contribution to the score.

However, since the Wall-Setter can force the Pawn-Pusher to visit every square at least once, I know that the sum of the lengths of all dead end branches and the true path must be the total size of the grid, 100 squares,

$$\sum_i l(b_i) + l_T = 100 \quad (2)$$

By employing this strategy, the Pawn-Pusher achieves a score of

$$s = 2(100 - l_T) + l_T - 1 \quad (3)$$

Here, I note that counterintuitively it is in the interests of the Pawn-Pusher for the true path to be as large as possible, since this eliminates wasted time spent going down dead ends. The smallest true path (and hence the largest score) is one which involves only moving up and to the right, traversing a total of 19 squares from (1, 1) to (10, 10). Therefore, by taking this strategy the Pawn-Pusher can ensure a score,

$$s \leq 2(81) + 18, \quad s \leq 180 \quad (4)$$

3. Wall-Setter Strategy

I now consider the Wall-Setter's strategy. Consider a smaller subset of the 10×10 grid with N total squares, all of which are exit points, and imagine the Pawn-Pusher has entered into this grid in such a way that all 3 unvisited sides of the square they are in are traversible. Imagine that the Wall-Setter chooses to allow the Pawn-Pusher to explore all 3 of these directions in turn before the Pawn-Pusher comes back to the first square with nowhere else to go but back out the way they came, as shown in Fig. 4. By doing this, the Wall-Setter forces a total score change,

$$\Delta s = 2(l(b_1) + l(b_2) + l(b_3)) \quad (5)$$

But the Wall-Setter can force the Pawn-Pusher to visit each square in this subgrid, therefore (excluding the entry point into the subgrid) $l(b_1) + l(b_2) + l(b_3) = N - 1$. This would be true regardless of the length of the dead end branches, or indeed the number of the branches. It is therefore of no benefit to the Wall-Setter to force the Pawn-Pusher down multiple branches spanning from the same grid point. It is not detrimental to the Wall-Setter, but there is no benefit in forcing this behaviour. For the Wall-Setter, the optimal strategy is to try to force the length of the true path l_T to be as small as possible. I do this as follows:

- On the first move, the Wall-Setter does nothing, and allows the Pawn-Pusher to take the first move. This allows the Pawn-Pusher to create a branch point at (1, 1).
- After the Pawn-Pusher makes the first move, the Wall-Setter places walls such that the number of unvisited exit points that the Pawn-Pusher can reach

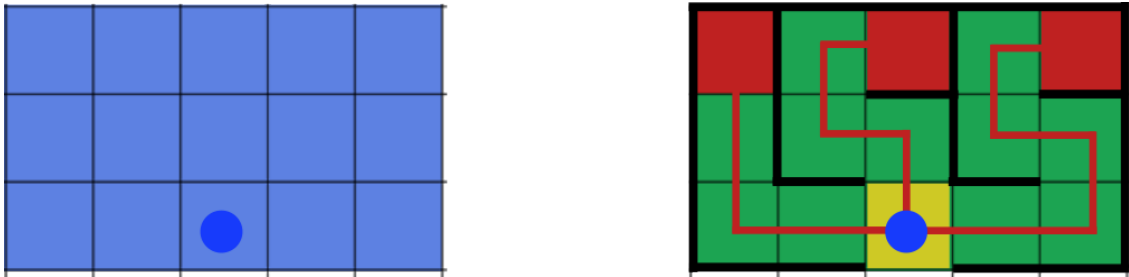


Figure 4: A way to force the Pawn-Pusher to visit every square in a subgrid.

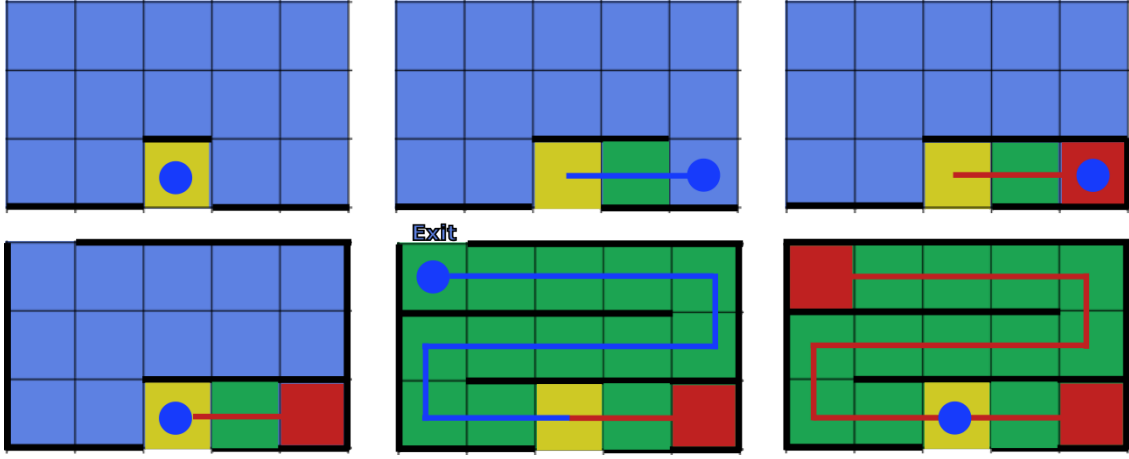


Figure 5: Ways for the Wall-Setter to force the Pawn-Pusher to visit every square.

without retracing its steps is always 1, dynamically creating a long branch in response to the Pawn-Pusher's moves.

- The Wall-Setter continues to place walls this way, always leaving an exit point open in front of the Pawn-Pusher at the end of the branch, always leaving a path back to (1, 1) and an unvisited route from (1, 1) to (10, 10), and always forcing the Pawn-Pusher to traverse every square along its path. When placing walls, the Wall-Setter must never create an unvisited branch without an exit point, which the Pawn-Pusher would be able to skip over without visiting. This approach is shown for a 3×5 subgrid in Fig. 5.
- The Wall-Setter lets the Pawn-Pusher walk all the way to the edge of the grid, either (10, 9) or (9, 10), and then blocks off access to the exit, forcing the Pawn-Pusher to traverse the full distance back from the dead end to (1, 1).

By keeping an exit point on both sides of the branch until the last possible moment, the Wall-Setter keeps the branch 'reversible'. i.e. if at any point the Pawn-Pusher decides to abandon its path before it reaches a dead end and turn around to the branch point they came from, the Wall-Setter can simply wait for the Pawn-Pusher to reach the square before an exit point, place a wall down to block the exit, and force the Pawn-Pusher to traverse back along the path a third time. This will always result in a higher score than if the Pawn-Pusher were to simply follow their path until it becomes a dead end.

Employing this strategy, the Wall-Setter can always ensure that if the Pawn-Pusher is playing optimally, each dead end branch is traversed exactly twice and the length of the true path $l_T = 19$. The smallest overall score that could be achieved with the Wall-Setter playing this strategy is therefore,

$$s \geq 2(81) - 18, \quad s \geq 180 \quad (6)$$

An example game played optimally at its end is shown in Fig. 6

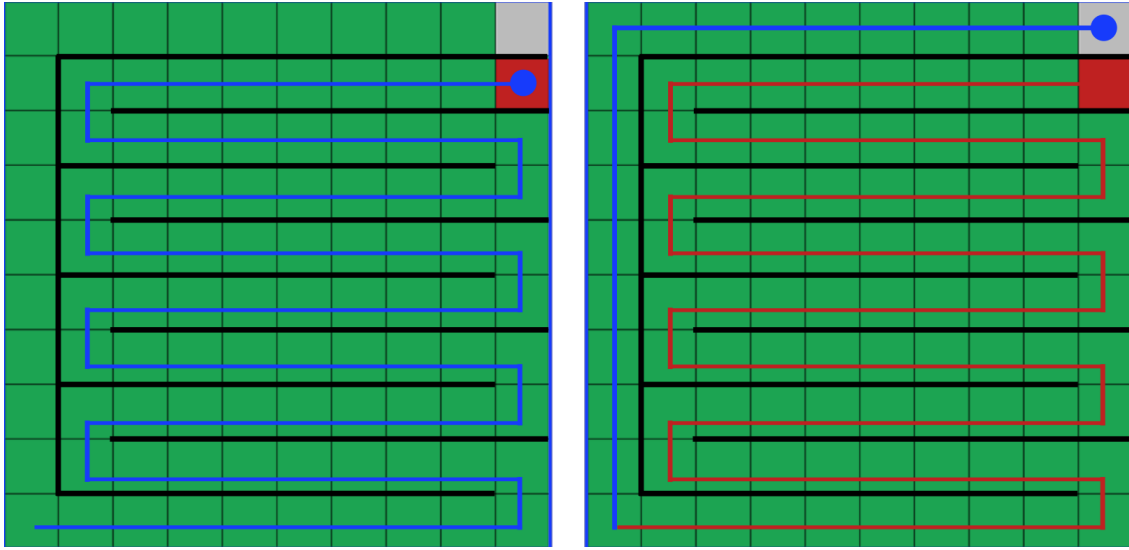


Figure 6: The final state of an optimally played game by both Wall-Setter and Pawn-Pusher with score $s = 180$.

4. Final Answer

Since the Pawn-Pusher can guarantee a score of at most 180 and the Wall-Setter can guarantee a score of at least 180,

Final Answer:

$s = 180$.

The Pawn-Pusher employs a depth-first search policy, traversing down a branch until it reaches a dead end before turning around, ensuring no square is revisited more than twice.

The Wall-Setter dynamically builds the maze piece by piece in response to the Pawn-Pusher's movements, forcing them to traverse a path of length 81 before closing the path off as a dead end, meaning the Pawn-Pusher has to traverse 81 steps back and then an additional 18 to (10,10).